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Analyzing Research Trends in Smart Construction Safety: A Topic Modeling Approach

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Abstract: The construction industry is increasingly embracing smart technologies to enhance safety, efficiency, and sustainability. Despite their potential, the practical integration of technologies such as digital twins, the Internet of Things (IoT), and big data into construction safety management systems remains insufficiently explored. This study investigates how these technologies can be effectively implemented to improve safety outcomes. A systematic review of the literature is conducted, culminating in the development of a conceptual framework for integrating smart technologies into safety systems. The study highlights the application of digital twins, the IoT, and big data for real-time monitoring, predictive risk management, and resource optimization. The findings reveal that these technologies significantly enhance construction site safety by proactively identifying hazards, reducing accidents, and improving resource allocation. Moreover, smart technologies contribute to environmental sustainability by optimizing energy use and lowering carbon emissions. This research underscores the dual benefits of technological integration, advancing both safety and sustainability objectives. While the study provides theoretical insights and practical implications, further empirical research across diverse construction environments is necessary to validate and refine the proposed framework.

Keywords: smart construction technologies; safety management; digital twins; IoT; big data; sustainability

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1. Introduction

The construction industry is a fundamental element of the global economy, driving economic growth and enabling infrastructure development. However, it continues to be one of the most hazardous sectors, with persistently high accident rates and frequent severe incidents [1]. Recognizing the persistent safety challenges in the construction industry, we emphasize the urgent need for robust safety management systems and innovative preventive measures to mitigate risks, safeguard worker health, and promote sustainable operations [2,3]. Beyond meeting regulatory and operational requirements, ensuring safety in construction represents both a moral imperative and an economic necessity that profoundly influences industry resilience and sustainability.

To address these critical concerns, our study examines recent advancements in smart construction safety, utilizing advanced technologies to transform traditional safety management practices. Emerging technologies such as the Internet of Things (IoT), artificial intelligence (AI), drones, wearable devices, and big data analytics have become integral to modern safety strategies [4–6]. These technologies enable real-time monitoring, predictive

risk assessment, and proactive safety interventions. For example, IoT sensors and AI-driven algorithms can identify hazardous conditions and issue timely alerts to prevent accidents. Our analysis underscores how these integrated systems significantly enhance operational awareness and risk mitigation on construction sites [7].

Despite advancements, implementing emerging technologies often remains fragmented, with few efforts to integrate them into comprehensive safety frameworks. Existing research predominantly focuses on standalone applications of individual technologies—such as drones for site inspections or IoT sensors for environmental monitoring—without fully addressing their synergistic interactions or cumulative impacts on safety outcomes [8,9]. This lack of integration underscores the necessity for a systematic analysis that examines how these technologies function both individually and collectively within safety management systems.

To bridge this gap, our research provides a detailed analysis of the intersection between advanced technologies and their practical applications in enhancing construction safety. Specifically, we focus on AI and machine learning technologies applied to worker safety management systems, a critical area with significant potential to reduce accidents and improve overall safety performance. By adopting an interdisciplinary perspective, we examine the interplay between technological capabilities and their practical implications on construction sites.

In response to the complexities and challenges associated with integrating diverse smart technologies, we deliberately narrow the research scope to focus on specific technologies and their targeted applications. By concentrating on artificial intelligence (AI) and its role in worker safety management systems, our study aims to offer a deeper and more coherent analysis of AI's transformative potential in construction safety. This focused approach facilitates a detailed exploration of AI's unique features and practical applications while ensuring logical consistency and depth throughout the study. By refining the research scope, we aim to provide actionable insights directly applicable to real-world construction scenarios and address the fragmentation often associated with broader studies. This strategic narrowing of focus enhances this study's academic rigor and practical relevance, addressing a critical gap in the literature and aligning with the urgent safety needs of the construction industry.

This research is guided by two primary objectives. First, we seek to identify and analyze trends in smart construction safety technologies. To achieve this, we systematically review the academic and industry literature, examining the deployment of technologies such as AI, the IoT, drones, and other innovations across various construction contexts. Particular emphasis is placed on understanding the interplay of these technologies when integrated into safety management systems. Real-world case studies are employed to illustrate the combined benefits of these integrated approaches [10–12].

Second, we evaluate the impact of technology integration on accident prevention and safety performance. While previous studies have often investigated specific technologies in isolation, our research adopts a holistic approach to assess the role of AI-driven safety solutions in developing comprehensive safety frameworks. We also examine significant challenges associated with technology adoption, including high costs, technical expertise requirements, and organizational resistance. Based on these findings, we propose practical recommendations to address these barriers and facilitate the effective implementation of advanced technologies in construction safety management [13,14].

By pursuing these objectives, our study contributes actionable insights into the synergistic application of advanced technologies in construction safety management. Specifically, we employ topic modeling as a robust analytical tool to systematically analyze and extract patterns from a vast corpus of the literature [15]. This method enables us to uncover latent

themes and evolving research trends in smart construction safety management, offering a comprehensive perspective that is difficult to achieve through traditional qualitative reviews. By leveraging the strengths of topic modeling—such as its scalability, objectivity, and ability to process large datasets—our study advances theoretical and practical understanding, providing a roadmap for the effective adoption and integration of smart safety technologies. Ultimately, we aim to support the construction industry in achieving enhanced worker safety, improved operational efficiency, and long-term sustainability [13,14].

2. Theoretical Background

2.1. Concept of Smart Construction Safety Management

Smart construction safety management integrates advanced technologies such as the Internet of Things (IoT), blockchain, and digital twins to enhance safety at construction sites and improve management efficiency [1]. These systems employ real-time data monitoring and risk analysis to make decisions that prevent accidents and anticipate potential risks before they occur. For instance, digital twin technology creates a virtual replica of construction sites, enabling the proactive identification and mitigation of hazards [1]. These technologies play a crucial role in strengthening safety measures through the use of real-time site data. Additionally, IoT devices and sensors continuously monitor environmental variables, thereby proactively addressing potential risks [2].

A fundamental aspect of smart construction safety management is the real-time monitoring of operations and data-driven decision-making processes. IoT devices and sensors collect continuous data on site conditions and worker behavior, which can be analyzed to identify risk factors early on. For example, smart inventory management systems utilize the IoT and cloud computing to monitor inventory levels and assess physical site conditions in real time, allowing for the rapid identification and prevention of potential hazards [2]. These systems are essential for pre-empting accidents and establishing robust frameworks for emergency response.

Accident management and response systems prioritize the rapid identification of accident causes and consequences, followed by the implementation of effective countermeasures. These systems analyze real-time data to quickly diagnose problems and propose appropriate solutions. For instance, variables such as temperature, humidity, and structural stability are monitored at accident sites to pinpoint root causes and assess associated risks [8]. Blockchain-based data management systems further enhance this process by securely storing accident-related data, ensuring real-time access for all stakeholders and maintaining the reliability and transparency of information [6].

Post-accident response procedures are increasingly automated and optimized. For example, automated systems can dispatch emergency rescue requests and provide optimal evacuation routes for on-site workers and managers [2]. Such systems reduce confusion immediately after an incident, facilitating swift and effective responses. They also empower workers and managers to monitor accident scenarios in real time, enabling informed and timely actions. These advanced accident response systems significantly enhance safety outcomes by ensuring efficient resolution processes, thereby fostering safer construction environments.

Smart construction safety management systems are essential not only for effective accident response but also for enhancing worker education and training practices [1]. An important application of these systems is applied to simulations powered by digital twin technology, which replicates real accident scenarios in virtual environments. This capability has been highlighted in text analysis results, where “digital twins” have emerged as a systematic theme, emphasizing their critical role in modern safety management practices. Such simulations provide workers with a risk-free platform to interact with potential hazards and practice response strategies in real time, significantly enhancing training

efficacy. By offering immersive virtual environments, digital twins improve workers' preparedness and contribute to safer on-site practices. These findings underscore the integration of advanced technologies like digital twins into training programs as a vital step in advancing construction safety management [1].

In addition to digital twin-based simulations, augmented reality (AR) and virtual reality (VR) have broadened the scope of construction safety training. These immersive technologies create environments that closely resemble actual construction sites, enabling workers to develop practical response skills beyond traditional theoretical training. AR and VR not only enhance workers' preparedness for potential accidents but also promote heightened awareness of safety protocols. This experiential learning strengthens the overall safety culture within the construction industry [2].

Key smart technologies, such as digital twins, the IoT, AR, and VR, play crucial roles in preventing accidents and facilitate rapid, efficient responses when incidents occur. Integrating these technologies into training programs significantly boosts safety awareness among workers by equipping them with the skills to manage various risk factors on construction sites. As workers become proficient in identifying and mitigating hazards, the likelihood of accidents decreases, improving the safety of construction environments [8,9].

Implementing smart construction safety management systems not only enhances accident response capabilities but also contributes to creating a sustainable safety management framework. These systems are instrumental in transforming the safety culture of the construction industry by providing tools for real-time risk assessment and decision-making. Moreover, accident management systems that leverage advanced technologies enable quick responses to incidents, thereby minimizing damage and fostering a proactive approach to safety management.

The integration of advanced technologies such as digital twins, the IoT, AR, and VR into construction safety management systems significantly enhances the theoretical understanding and practical preparedness of workers. Working synergistically, these technologies prevent accidents, improve safety outcomes, and foster a culture of safety within the industry. As the construction sector continues to adopt and implement these technologies, the prospects for establishing more efficient and sustainable safety management systems increase. This technological advancement is essential for the ongoing improvement of safety standards and the promotion of long-term cultural change within the industry.

2.2. Cases and Effects of Construction Safety Management Utilizing Smart Technology

Recent advancements in smart technology have significantly impacted construction safety management, positioning technologies such as the Internet of Things (IoT), blockchain, digital twins, and artificial intelligence (AI) as foundational tools for improving safety standards in the construction industry. These technologies facilitate the prediction of potential risks at construction sites and enable quick responses to accidents, ultimately enhancing the safety environment on-site.

For instance, IoT sensors are widely used to monitor environmental factors such as temperature, humidity, and structural stability at construction sites. By providing real-time data, these sensors aid in the early detection of risk factors, allowing for timely interventions. The ability to continuously monitor these factors supports proactive risk management, reduces the likelihood of accidents, and improves safety outcomes [8]. In addition, blockchain technology has been implemented in data management systems to securely record accident-related information. These systems provide real-time access to relevant stakeholders, ensuring transparency and trust in the data, which is crucial for effective decision-making during safety management processes [10].

One of the most innovative technologies in construction safety management is digital twin technology, which creates virtual replicas of construction sites. This technology enables the real-time monitoring and simulation of various scenarios, allowing for the prediction of accident scenarios and the identification of risk factors before they manifest in reality. For example, in a specific construction project, digital twins were utilized to simulate virtual scenarios in real time, thereby guiding the development of safety protocols and mitigating potential hazards based on the simulation results. This capability has established digital twin technology as a crucial tool for maximizing construction site safety and preventing accidents.

Furthermore, the integration of AI and machine learning into smart construction safety systems plays a pivotal role in preventing accidents. These systems analyze vast amounts of data in real time to detect patterns and identify risks that may lead to accidents. AI can also evaluate worker behavior, automatically identifying unsafe practices and issuing real-time alerts to avert accidents before they occur. Additionally, these AI systems can manage post-incident responses, significantly decreasing response times and improving the efficiency of on-site actions after an incident [12]. Through these capabilities, AI-driven systems contribute to reducing accidents and mitigating damage by ensuring rapid, well-coordinated responses.

The integration of the IoT, blockchain, digital twins, and AI into construction safety management systems is proving essential for not only predicting risks and preventing accidents but also for enabling swift, effective responses when accidents occur. These technologies collectively play a crucial role in minimizing construction site accidents, enhancing safety practices, and optimizing emergency response times.

Smart construction safety management systems have proven to be highly effective not only in preventing accidents but also in enhancing education and training programs. These systems, particularly those that incorporate digital twin technology, augmented reality (AR), and virtual reality (VR), simulate real-world accident scenarios within virtual environments, thus providing workers with real-time response training [9]. Unlike traditional theoretical approaches, these simulations closely replicate actual conditions, allowing workers to improve their response capabilities in scenarios that mirror real-life situations they may encounter on construction sites.

AR and VR technologies are particularly instrumental in helping workers develop the skills necessary to quickly identify and react to hazardous situations without having to directly experience an accident. These training methods have shown considerable promise in reducing errors and minimizing the likelihood of accidents on site. By immersing workers in a controlled, virtual environment, these technologies enhance their decision-making abilities, enabling them to act swiftly and accurately under pressure.

In conclusion, the integration of advanced technologies—such as the Internet of Things (IoT), blockchain, digital twins, artificial intelligence (AI), AR, and VR—into construction safety management systems plays a pivotal role in transforming safety practices across the industry. These technologies work synergistically to enable proactive accident prevention and facilitate rapid, efficient responses to incidents, thereby significantly enhancing safety outcomes on construction sites. Moreover, as these technologies continue to evolve, the construction industry is poised to witness even greater improvements in safety standards, fostering the continuous innovation of safety culture within the sector [13].

2.3. Importance of Research for Technological Advancement and Sustainable Development

Research is a fundamental driver of technological innovation and sustainability, enabling the identification of current challenges, the development of innovative solutions, and the refinement of existing technologies. Through continuous research efforts, advance-

ments can be fostered that not only improve efficiency but also support the long-term well-being of the environment and society. Smart construction safety management systems that integrate technologies such as the Internet of Things (IoT), artificial intelligence (AI), blockchain, and digital twins offer transformative solutions for enhancing safety at construction sites. However, their effective deployment and sustainable evolution depend on persistent research and technological advancements [1,3].

While IoT sensors enhance accident prevention through real-time data collection and analysis, unresolved issues such as data accuracy, security, and privacy protection persist. Enhancing the reliability of these systems in complex data environments remains a critical research priority. Similarly, while blockchain technology promotes robust data transparency and security, it struggles with slow data processing speeds and storage inefficiencies. Addressing these challenges is crucial to reducing operational costs and improving scalability [8]. Digital twin technology, which virtualizes construction sites for safety simulations, still needs advancements to allow for the more accurate modeling and better integration of real-time data [1,5]. Tailored technological solutions, supported by experimental research, are vital for adapting to diverse construction environments [4,7].

Blockchain technology, known for its data transparency and security, requires improvements in data processing speeds and storage capacities to be practically effective [8]. Overcoming these challenges can lead to reduced operational costs and enhanced scalability. Likewise, digital twin technology needs further development to achieve more accurate modeling and effective real-time data integration [1,5]. Developing customized technologies through experimental research, suited to various construction site conditions, is crucial for addressing these issues [4,7].

To maintain the advantages of smart construction safety management systems, it is important to minimize their environmental and economic impacts [2,9]. Despite the proficiency of IoT and AI technologies in handling large data volumes, their substantial energy requirements pose a challenge to environmental sustainability. Enhancing energy efficiency and optimizing operational systems are essential steps. Blockchain technology, notorious for its high resource usage, needs sustainable utilization strategies [10,14]. Moreover, increasing the interoperability among these technologies and creating integrated networks are key for seamless functionality in actual construction settings.

AR (augmented reality)- and VR (virtual reality)-based training systems, designed to simulate accident scenarios in virtual settings, need to evolve to more accurately reflect the complexities of real construction sites. Additionally, IoT sensors and AI systems must undergo thorough testing to confirm their effectiveness across various field conditions [11–13]. This comprehensive research agenda is crucial for promoting a safety culture and enhancing accident prevention in the construction industry. By addressing these challenges, smart construction safety management systems can lead to technological advancements, improved safety outcomes, and the adoption of sustainable safety practices [4,10].

2.4. Smart Construction Safety: A Proactive Risk Management Model

According to previous studies, we actively developed a systematic framework for addressing the challenges of smart construction safety management. By leveraging risk management theory, we identify, assess, and propose strategies to control and monitor potential risks that may arise within complex systems. Risk management theory, guided by international standards such as ISO 31000 [15], provides a comprehensive approach to mitigating risks that hinder organizational goals. Drawing on the work of Sousa et al. [16], who introduced a risk management model for occupational safety and health in the construction industry, we emphasize key processes such as understanding, analyzing, and modeling accidents. This model serves as a critical foundation for real-time risk identification and

management in technologically advanced environments like smart construction, enabling simultaneous improvements in safety and efficiency.

In developing our approach, we integrated advanced technologies including the IoT, AI, big data, digital twins, blockchain, AR, and VR to establish a proactive risk management framework. IoT sensors enable the real-time monitoring of environmental variables, facilitating early hazard detection, while AI analyzes behavioral patterns to prevent unsafe practices. Digital twins simulate scenarios to predict and resolve risks before they manifest, and blockchain enhances data transparency to support reliable decision-making. AR and VR technologies revolutionize worker training by creating immersive simulations of risk scenarios, enhancing preparedness. These innovations collectively reinforce our framework, equipping it to effectively mitigate accidents, optimize responses, and ensure continuous safety improvements.

Our framework builds on the foundational processes of risk management theory. We actively engage in risk identification, pinpointing potential threats across organizational activities and technological systems. Through risk assessment, we evaluate the likelihood and severity of each identified risk, prioritizing those requiring immediate attention. In the risk control phase, we develop strategies to address risks, including avoidance, mitigation, or acceptance, depending on their nature. Finally, we incorporate robust mechanisms for risk monitoring and review, ensuring ongoing evaluation and adaptation to evolving conditions. This structured approach is particularly essential for the complex systems of smart construction, where advanced tools introduce opportunities and novel risks.

Drawing insights from Sousa et al. [16], we adopted a quantitative methodology for assessing occupational safety risks, aligned with ISO 31000 [15]. This approach emphasizes systematic evaluation processes, including accident analysis and modeling, to establish clear risk criteria and thresholds. Such a methodology not only underpins effective risk management but also enables its application in technologically advanced contexts like smart construction, where real-time adaptability is critical. Our proactive adoption of this model facilitates the identification and mitigation of risks in ways that enhance operational safety and efficiency.

As part of our study, we focus on integrating advanced technologies to transform traditional construction workflows while addressing the unique challenges posed by these innovations. Technologies such as the IoT, AI, big data, and digital twins allow us to propose methods that increase productivity and safety while maintaining alignment with global sustainability goals. By embracing these technologies, we pave the way for creating environmentally and socially sustainable construction environments. Our proactive approach ensures that emerging risks are systematically managed, contributing to a resilient and sustainable construction sector.

To achieve our research objectives, we establish three strategic frameworks. The first, "Technological Convergence for Enhancing Construction Efficiency and Safety", focuses on integrating cutting-edge technologies with traditional construction workflows to improve productivity and safety. This framework addresses both existing and emerging risks through innovative solutions. The second, "Sustainable Smart Environments through Technological Integration", examines the role of smart technologies in promoting sustainability within the construction industry. By aligning with global sustainability objectives, this framework demonstrates how digital innovations can support the creation of sustainable construction environments. Finally, the "Worker-Centered Safety Management Systems and Technological Support" framework emphasizes the importance of real-time monitoring and predictive analytics in enhancing worker safety. It proposes adaptive strategies to optimize safety and efficiency, fostering safer and more effective work environments.

Through these frameworks, we actively bridge the gaps between technological innovation, sustainability principles, and worker-centered safety management. By addressing the interconnected challenges of smart construction, we provide a robust theoretical and practical foundation for future advancements. This proactive and integrated approach not only contributes to the understanding of smart construction safety management but also establishes a transformative model for enhancing safety, efficiency, and sustainability in the construction industry.

3. Research Methodology

Topic modeling is a computational technique widely used to analyze large datasets by identifying latent topics and patterns within textual data, providing insights into the underlying structure of the information [15–18]. Unlike surveys or interviews, which rely on directly collected data and subjective interpretations, topic modeling allows for the objective analysis of extensive textual corpora, making it particularly suitable for identifying research trends and thematic structures in a given field. A key parameter in topic modeling is the number of topics, which is often determined subjectively by the researcher, as there is no definitive statistical method for setting this value [18,19]. While recent approaches, such as complexity-based data modeling [19] or using harmonic mean values to optimize the model's coherence and sample collection, offer computational solutions, they can still fall short in capturing the semantic nuances of the topics. Therefore, the most commonly adopted method relies on the researcher's domain expertise and objectives to define the number of topics. By leveraging this method, this study systematically analyzes the corpus of the smart construction safety management literature, showcasing the clear distinction between computational text analysis and traditional qualitative approaches.

3.1. Selection of Research Subjects

This study investigates the domain of smart construction safety, focusing on critical areas such as safety management, educational methodologies, and training practices that leverage emerging smart technologies. This research places particular emphasis on the technologies driving innovation in this field, including the Internet of Things (IoT), wearable devices, big data, artificial intelligence (AI), drones, robotics, augmented reality (AR), virtual reality (VR), building information modeling (BIM), and digital twins.

To ensure a focused and relevant body of research, the scope was refined to include only English-language sources. Keywords such as "smart technology", "IoT", "wearable device", "big data", "artificial intelligence", "drone", "robot", "augmented reality", "virtual reality", "BIM", and "digital twin" were utilized. These terms were selectively chosen to align with this study's objectives and to ensure a comprehensive examination of the subjects.

This literature review was conducted using premier academic databases such as Scopus, Web of Science, ScienceDirect, and Google Scholar, known for their extensive collections of high-quality and current research publications. These databases were selected to fulfill this study's goals by providing multidisciplinary access to the latest developments in smart construction safety. Boolean operators (e.g., AND, OR) were strategically used to refine the search process. The keywords were systematically combined based on their relevance in the topics, titles, abstracts, and keywords of the literature.

Table 1 outlines the detailed search strategy implemented in this study. It illustrates how Boolean operators and specific keyword combinations were used to ensure a targeted and comprehensive review of the extant research. This methodical approach aimed to encompass a broad spectrum of insights and findings, thereby laying a solid foundation for understanding both advancements and challenges in the field of smart construction safety.

Table 1. Search strategy in this study.

Title 1	Details of Search Criteria
Topic	"construction management"
Title	("construction safety" OR "worker safety" OR "industrial safety") AND ("IoT" OR "AI" OR "BIM" OR "wearable devices" OR "augmented reality" OR "digital twin")
Abstract	("construction site" OR "construction industry") AND ("smart technology" OR "IoT" OR "wearable devices" OR "AI" OR "big data")
Keywords	("construction safety" OR "worker safety" OR "industrial safety") AND ("smart construction" OR "IoT" OR "BIM" OR "AI" OR "robotics")

This study aimed to provide comprehensive insights into smart construction safety, focusing on the latest technologies and pertinent case studies. The research spanned a deliberately broad period to encapsulate the most recent developments in the field. The literature search was implemented in two phases: the first was from 1 October to 12 October 2024 and the second was from 4 November to 9 November 2024. Searches were confined to English-language publications to align with international research standards. Through this meticulous approach, the relevant literature was identified and assessed. Initially, a total of 1360 papers were collected. Throughout the selection phase, the papers were screened based on full-text accessibility and relevance to the research topic, committing to data integrity by excluding duplicate publications, such as theses published both as journal articles and conference presentations, with a focus on journal articles.

During the review process, various exclusion criteria were methodically employed to refine the document selection. Initially, 393 papers were excluded due to a lack of professionalism or relevance to this study's goals. This step included verifying the credibility of publishers to ensure consistency with the research theme. Furthermore, 66 duplicate documents were identified and eliminated to preserve the dataset's uniqueness and integrity. Additional refinement led to the exclusion of 85 papers unrelated to industrial safety and 90 papers that did not concentrate on smart construction safety. These measures guaranteed a high-quality and targeted dataset for analysis. After the initial screening, 901 documents remained. A subsequent review further narrowed this number to 726 documents. During this literature analysis, four new datasets were identified and added, resulting in a final dataset of 730 documents for analysis.

In this study, we actively applied the PICO process as a systematic literature review methodology to analyze trends in smart construction safety research. The PICO process provided a structured framework for defining research questions and efficiently identifying the relevant literature. It comprised four components: Population/Problem, Intervention, Comparison, and Outcome [20,21]. During the Population/Problem stage, we identified research subjects and issues related to smart construction safety management, such as "construction site", "construction industry", "worker safety", "industrial safety", and "work environment." Based on these, we defined search terms including "construction site", "construction industry", "worker safety", "industrial safety", and "hazard prevention."

In the Intervention phase, we focused on identifying interventions and key technologies relevant to smart construction safety. Search terms such as "smart technology", "IoT", "wearable device", "big data", "artificial intelligence", "drone", "robot", "augmented reality", "virtual reality", "BIM", and "digital twin" were employed. For the Comparison stage, we introduced terms like "traditional safety management" and "conventional construction safety methods" to compare modern technologies with traditional safety management approaches. Finally, in the Outcome phase, we selected terms such as "accident prevention", "improved safety", "efficiency", "risk management", "technology adoption", and "cost efficiency" to focus on key results and measurement criteria.

By employing this structured PICO process, we systematically conducted a comprehensive literature search to identify pivotal studies related to smart construction safety management. We evaluated the retrieved literature for relevance and alignment with our research objectives. It enabled us to establish a robust theoretical foundation, clarify the interconnection between smart technologies and construction safety, and propose research directions for effective risk management and accident prevention. Through this process, we ensured that our study contributes meaningfully to advancing the understanding of smart construction safety practices.

The selected documents were transformed into unstructured textual data for text mining analysis. The Title–Keywords–Abstract (TKA) method was utilized to systematically organize the data. Figure 1 presents a flowchart that outlines the data selection process for this literature review. This structured and systematic approach ensured a robust dataset, giving a strong foundation for this study's emphasis on integrating smart technologies into construction safety systems.

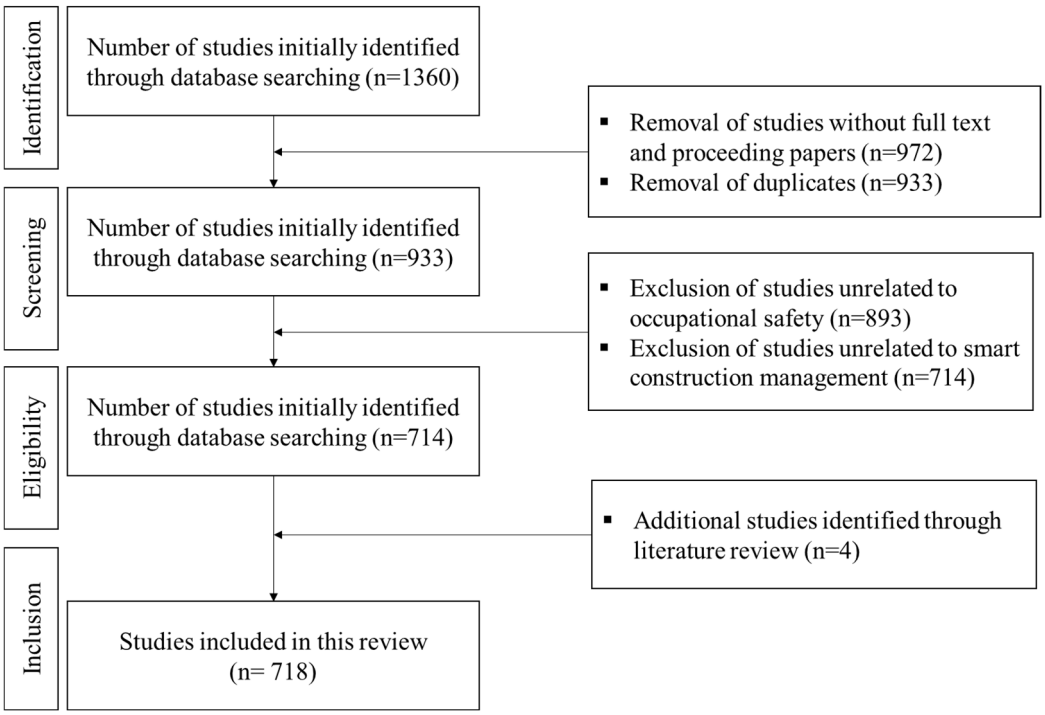


Figure 1. Flowchart of article selection in this systematic literature review.

3.2. Data Analysis Method

In this section, we detail the methodologies employed to analyze the data gathered, ensuring that the findings were robust and reliable, thereby supporting valid conclusions. This study utilized topic modeling and data visualization techniques to explore research trends within the field of smart construction safety management. Text mining was the principal method for deriving meaningful insights from an extensive collection of documents, proving particularly adept at analyzing varied sources such as academic journals, reports, and articles [18,19]. Through the identification of key keywords and themes, text mining elucidates major research areas, forecasts interrelationships, and explores the developmental possibilities of different topics. This method reduces subjective biases from researchers and guarantees precise classification of information, even when papers cover multiple domains [18].

The analysis commenced with the extraction of keywords from the documents collected, followed by a frequency analysis and topic modeling to ascertain the importance of each keyword. The relationships among key terms were examined through centrality

analysis, and the findings were visually depicted using a word cloud. For this analysis, R software (version 3.6.3) was used, along with various packages such as NLP, tm (for English texts), SnowballC (for root word extraction), stringr, dplyr (for string processing and visualization), and lda (for corpus analysis).

During preprocessing, stop words like "proposed", "methods", "related", and "factors", as well as special characters and numbers, were eliminated to improve data clarity. The number of topics for modeling was determined by a blend of computational methods and the judgment of the researchers. Although modern computational techniques, such as maximizing the harmonic mean, facilitate the optimization of topic numbers, the interpretation of these topics remains heavily dependent on the expertise of researchers [18,19]. Adjusting the number of topics to match the depth and scope of research is essential to accurately mirror the objectives of a study [22–24].

The primary goal of this study was to identify the core research areas within smart construction safety management. Key topics included smart technology, the IoT, AI, and robotics. The analysis covered broad research categories while conducting in-depth examinations specific to each keyword. Using the TF-IDF (term frequency–inverse document frequency) weighting model, the frequency and significance of keywords were assessed, facilitating the identification of primary topics. These topics were then visually represented to offer an intuitive grasp of the research trends in the field of smart construction safety management.

In this study, we provide a detailed explanation of the centrality, TF (term frequency), and TF-IDF (term frequency–inverse document frequency) methods employed during the data analysis process to enhance the transparency and rigor of our research methodology. Centrality analysis was used to assess the importance of specific words or topics within the dataset. The centrality index allowed us to examine correlations between research keywords and identify the most influential topics within the field of smart construction safety management.

Centrality, commonly used in network analysis to measure the importance of nodes within a network [18], was adapted in this study to focus on the relative significance of words in the dataset rather than their relationships within a network. Specifically, centrality was determined by calculating the ratio of each word's frequency to the total frequency of all words. This approach quantitatively reflected how prominently a given word appeared within a text, providing a focused evaluation of its importance.

Calculating centrality involved a series of methodical steps to ensure accurate assessment. First, the textual data underwent preprocessing, where irrelevant elements such as stop words, special characters, and redundant terms were removed to enhance clarity and emphasize meaningful content. Following this, the frequency of each word in the dataset was calculated, offering a foundational understanding of the distribution of terms. Lastly, the ratio of each word's frequency to the cumulative frequency of all words in the dataset was computed. This calculation facilitated a quantitative evaluation of the relative importance of words, which proved essential in identifying dominant topics and research trends within this study.

The *TF* and *TF-IDF* are important analysis techniques in text mining, which are used to measure the frequency of words in textual data and evaluate the relative importance of the words in a document [19]. The TF is an indicator that measures how frequently a specific word appears in a document. It is calculated by dividing the frequency of a word, "t", by the total number of words in document, "d", and is expressed by the following Formula (1).

$$TF(t,\ d) = \frac{f(t,\ d)}{N_d} \tag{1}$$

where we have the following:

- $f\ (t, d)$ is the number of times the word "t" appears in the document "d".
- N_d is the total number of words in the document "d".

The *TF-IDF* is a measure of the importance of words, giving higher weight to words that appear frequently in a document but rarely in the entire document set [19]. It is calculated by combining the TF and IDF (inverse document frequency) and is expressed by the following Formula (2).

$$TF - IDF\ (t,\ d,\ D) = TF\ (t,\ d) \times IDF\ (t,\ D) \tag{2}$$

where the *IDF* is defined as

$$IDF(t,\ D) = \log \frac{|D|}{1 + |\{\mathrm{d} \in \mathrm{D} : \mathrm{t} \in \mathrm{d}\}|} \tag{3}$$

where we have the following:

- $|D|$ is the total number of documents.
- $|\{\mathrm{d} \in \mathrm{D} : \mathrm{t} \in \mathrm{d}\}|$ is the number of documents containing the word "t".

The *TF* and *TF-IDF* are pivotal analytical techniques in text mining, employed to assess the frequency of words in textual data and evaluate their relative importance within a document. For instance, if a specific keyword appears frequently but is commonly found across most documents, its significance is considered low. In contrast, a keyword that appears frequently in only some documents is given a higher weight. These computations were conducted utilizing the NLP and tm packages in R software, with R version 3.6.3 employed for the analysis.

Topic modeling analysis utilized the Latent Dirichlet Allocation (LDA) algorithm to identify key themes from the textual data by examining the distribution of topics within the documents. LDA operates under the assumption that each document is composed of a probabilistic mixture of multiple topics, and it identifies the primary topics and their respective distributions across the dataset [19,22]. In this study, LDA was instrumental in analyzing and interpreting the relationships and relevance among the identified topics, providing a structured understanding of the research domain.

The determination of the optimal number of topics was achieved through a combination of computational methods, such as harmonic mean optimization, and the researchers' domain expertise to ensure meaningful results. Furthermore, data visualization tools, including word clouds, were employed to represent the associations between keywords and topics visually, enhancing the interpretability of the analysis. These methodologies were rigorously applied to derive insights from the data and contribute to a comprehensive exploration of the research themes.

4. Results

4.1. Research Trends by Year

The analysis of trends in smart construction safety research shows a steady increase in the number of related publications over time. A notable rise in these publications is observed after 2017, attributed to the construction industry's proactive integration of technologies from the Fourth Industrial Revolution [13,23,24]. Research peaked in 2019, likely due to the growing recognition of the importance of incorporating smart technologies within safety management systems. This concept has become well-established in academic discussions and industrial sectors [25–29]. Figure 2 displays these research trends.

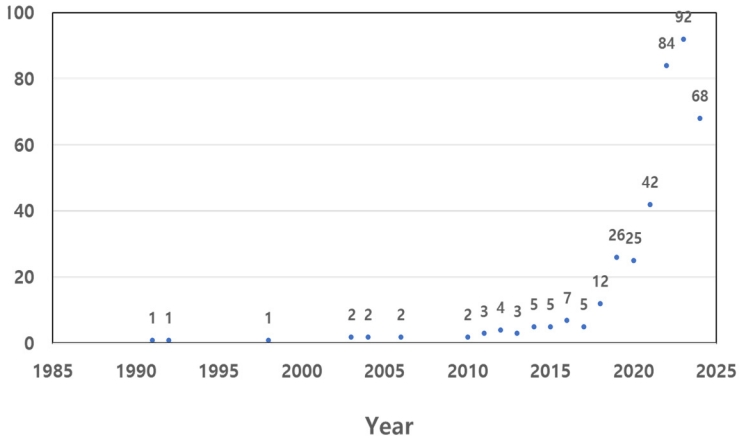


Figure 2. Annual trend of papers related to smart construction safety management.

The increase in research on smart construction safety technology is closely tied to broader societal changes, including an aging workforce and a declining working-age population. Initially, this technological area received little attention. However, as the necessity to employ advanced technologies—such as the IoT, digital twins, big data, and AI—to address safety issues for vulnerable groups like elderly workers became more apparent, research efforts significantly expanded. The notable rise in published works since 2017 reflects this societal shift and heightened awareness [30–32].

In the 2020s, the volume of research papers on smart construction safety technology has consistently increased, peaking at 84 papers in 2022 and 92 in 2023. This upward trend suggests that smart construction safety technology is shifting from theoretical research to practical implementation within the industry. Recent studies have increasingly focused on key aspects of smart construction safety, such as real-time monitoring, data-driven decision-making systems, and the use of advanced technologies for accident prevention [32–35]. These advancements indicate that research in smart construction safety is progressing to address both the enhancement of productivity and the improvement of worker safety in the construction sector.

In conclusion, the advancements in research concerning smart construction safety underscore the convergence of societal shifts and industrial needs. The escalating scholarly interest in this domain underscores the importance of smart construction safety as a pivotal area of study, poised to lead digital transformation and foster safety innovations in the construction industry going forward.

4.2. Text Frequency Analysis Results

Our text frequency analysis of the smart construction safety research indicated that the keywords “construction”, “smart”, and “technologies” were the most frequently mentioned, with “construction” being particularly prominent. This underscores their pivotal role in research that integrates digital technologies such as the IoT, AI, and automation with traditional construction methods. Such integration is catalyzing profound changes in construction site operations and management practices.

The terms “management” and “systems” also ranked highly, highlighting their significance in the realm of smart construction. These results underscore the necessity for efficient resource allocation, enhanced project performance, and the development of integrated management systems, all contributing to improved operational efficiency and safety. The critical nature of implementing smart systems for the management of complex projects was particularly noted. Additionally, terms associated with data usage like “big data”, “digital twin”, and “information” were frequently observed, emphasizing the importance

of data-driven technologies in enhancing real-time monitoring and predictive analytics at construction sites. These technologies are pivotal in boosting both safety and efficiency, thus positioning data-centric smart construction management as a crucial research area.

Furthermore, the prevalence of keywords such as "applications", "buildings", and "cities" suggests that the scope of smart construction research is broadening from traditional projects to encompass smart buildings and urban development. This broadening indicates that smart technologies are integral to fostering sustainable construction practices and underscores the increasing necessity for their integration into sustainable urban planning.

We observed a high frequency of terms related to safety and security, encompassing 'safety', 'monitoring', and 'security', underscoring an essential focus on worker protection and risk management in smart construction safety research. Primarily, the early detection of risks through real-time monitoring technologies and addressing cybersecurity have become key research areas to ensure safety in smart construction. Table 2 displays the results of the word frequency analysis.

Table 2. Text frequency analysis results.

No	Word	Frequency	Centrality	No	Word	Frequency	Centrality
1	Construction	1959	0.1049	19	Network	255	0.0137
2	Smart	1774	0.0950	20	SmartThings	224	0.0120
3	Technologies	1644	0.0880	21	Environment	221	0.0118
4	Management	1531	0.0820	22	Monitoring	217	0.0116
5	Systems	1246	0.0667	23	IoT	202	0.0108
6	Big Data	1192	0.0638	24	Security	185	0.0099
7	Qualitative Research	1117	0.0598	25	platform	184	0.0099
8	information	716	0.0383	26	learning	169	0.0090
9	buildings	640	0.0343	27	infrastructure	167	0.0089
10	cities	559	0.0299	28	practices	160	0.0086
11	applications	554	0.0297	29	knowledge	155	0.0083
12	digital twin	542	0.0290	30	evaluation	148	0.0079
13	safety	523	0.0280	31	cloud	145	0.0078
14	blockchain	373	0.0200	32	workers	144	0.0077
15	processes	368	0.0197	33	decision	138	0.0074
16	sites	359	0.0192	34	grid	138	0.0074
17	BIM	302	0.0162	35	support	137	0.0073
18	intelligent	287	0.0154				

Figure 3 visualizes the results of the text frequency analysis as a word cloud, clearly identifying the top 100 words and providing an immediate visualization of the core keywords in smart construction safety research.

Our analysis confirms that research in smart construction safety is advancing by integrating digital technology with traditional construction management practices. We recognize this convergence as essential for tackling the dual challenges of enhancing safety and efficiency in construction settings. Through this study, we pinpoint three primary research directions: the development of data-driven real-time monitoring and predictive technologies; the exploration of sustainable applications in smart buildings and urban planning; and the implementation of worker-focused safety management systems as pivotal in construction innovation.

In the first research topic, 'Technological Convergence for Enhancing Construction Efficiency and Safety', we explore how advanced digital technologies such as the IoT, AI, big data, and digital twins can be systematically integrated into construction processes. We analyze how these technologies can bridge the gap between traditional workflows and modern innovations, simultaneously improving productivity and safety. Our research

highlights practical solutions and strategic frameworks that demonstrate the transformative impact of cutting-edge technology on the construction industry.

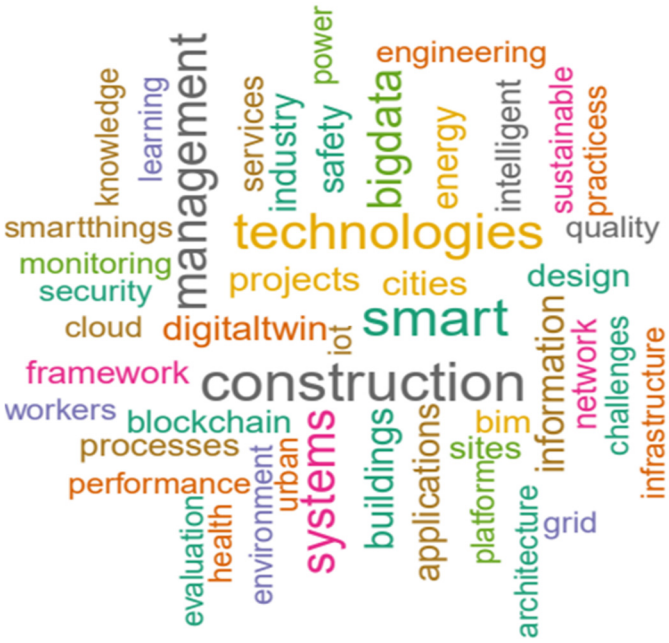


Figure 3. Text frequency analysis results (word cloud).

For the second research topic, 'Sustainable Smart Environments Through Technological Integration', we examine how smart technology can foster sustainable development within the construction sector. Specifically, we focus on applications for smart buildings and urban planning, investigating how digital innovations align with global sustainability goals. Our findings highlight the role of smart technology in enhancing environmental and social sustainability, offering actionable insights into its implementation in construction practices.

The third research topic, "Worker-Centered Safety Management Systems and Technological Support", enables us to prioritize the development of systems designed to enhance worker safety and improve on-site working conditions. We formulate specific recommendations for incorporating real-time monitoring and predictive analytics into safety management systems to proactively identify and mitigate risks. Furthermore, we suggest strategies integrating ergonomic design and adaptive technologies to forge safer and more efficient work environments, stressing the critical role of worker-focused solutions in advancing safety outcomes.

These three research topics embody the core directions of our study, underscoring the importance of a holistic approach that merges technological innovation, sustainability principles, and worker-centered safety practices. By addressing these interconnected domains, our research not only seeks to enhance efficiency and safety but also establishes the groundwork for a sustainable and technologically sophisticated construction environment.

4.3. Topic Modeling Analysis Results

In this study, we employed the topic modeling technique to pinpoint core topics within the smart construction industry safety management system. Through this analysis, we evaluated the importance of technological integration, sustainability, and worker safety management. The topic modeling analysis delineated three principal categories, each representing a vital element in enhancing the safety management system by implementing smart technologies in the construction industry. Table 3 displays the comprehensive results of the topic modeling analysis.

Table 3. TF-IDF analysis of key terms by research topic.

Technological Convergence for Enhancing Construction Efficiency and Safety							
No.	**Word**	**TF**	**TF-IDF**	**No.**	**Word**	**TF**	**TF-IDF**
1	construction	1959	1684.1587	16	IoT	202	716.8015
2	technologies	1644	4250.9095	17	sustainable	170	548.3773
3	management	1531	1677.7866	18	cloud	145	495.1737
4	big data	1192	4022.0091	19	support	137	507.2662
5	qualitative research	1117	4135.8855	20	adoption	127	485.1979
6	information	716	1731.7586	21	mobile	94	371.6748
7	buildings	640	1354.0529	22	algorithm	91	217.3400
8	applications	554	1314.9394	23	recognition	89	284.0744
9	digitaltwin	542	1748.3558	24	devices	88	214.2062
10	blockchain	373	869.2260	25	solution	80	145.7888
11	processes	368	1482.3390	26	equipment	62	181.6279
12	BIM	302	877.0579	27	dynamic	61	157.7284
13	network	255	767.4290	28	emerging	57	156.3451
14	performance	229	835.5307	29	risks	52	159.4676
15	monitoring	217	699.9875	30	wireless	52	165.9761
Sustainable Smart Environments through Technological Integration							
No.	**Word**	**TF**	**TF-IDF**	**No.**	**Word**	**TF**	**TF-IDF**
1	construction	1959	1684.1587	16	design	350	429.4740
2	smart	1774	445.1376	17	BIM	302	877.0579
3	technologies	1644	4250.9095	18	services	299	527.6545
4	management	1531	1677.7866	19	intelligent	287	484.3899
5	systems	1246	836.6468	20	framework	260	478.2581
6	big data	1192	4022.0091	21	network	255	767.4290
7	information	716	1731.7586	22	performance	229	835.5307
8	buildings	640	1354.0529	23	SmartThings	224	432.2316
9	cities	559	709.9493	24	environment	221	310.6506
10	applications	554	1314.9394	25	monitoring	217	429.2095
11	digitaltwin	542	1748.3558	26	carbon	132	400.9759
12	safety	523	939.9638	27	computing	132	329.8284
13	energy	479	889.4293	28	operation	125	200.3265
14	blockchain	373	869.2260	29	production	125	306.2387
15	processes	368	1482.3390	30	risks	52	159.4676
Worker-Centered Safety Management Systems and Technological Support							
No.	**Word**	**TF**	**TF-IDF**	**No.**	**Word**	**TF**	**TF-IDF**
1	Safety	523	939.9638	16	Education	83	283.4442
2	Design	350	429.4740	17	Collaboration	82	223.1912
3	Environment	221	310.6506	18	Human	73	201.8014
4	Monitoring	217	429.2095	19	Virtual Reality	72	188.8382
5	Engineering	161	372.9369	20	Requirements	70	177.2821
6	Knowledge	155	348.6321	21	Stakeholders	67	178.3017
7	Evaluation	148	357.9613	22	Assessment	61	203.4318
8	workers	144	450.3324	23	initial	59	174.3720
9	decision-making	138	274.3267	24	hazards	52	159.4676
10	assistance	137	507.2662	25	crisis	50	201.4048
11	safety	107	269.1593	26	education	45	161.8791
12	enhancement	94	272.9915	27	comprehension	44	126.6968
13	strategy	90	263.6535	28	workplace	43	173.2081
14	assets	88	247.1792	29	reaction	41	125.7340
15	instruments	84	235.9438	30	obstacles	40	152.8182

Topic 1 (Technological Convergence for Enhancing Construction Efficiency and Safety) investigates how the integration of advanced technologies can significantly enhance both efficiency and safety within the context of smart construction. Insights derived from the TF-IDF analysis identify several pivotal terms that highlight the role of technological convergence in optimizing construction processes.

The analysis underscores key terms, including "construction" (TF-IDF: 1684.16), "technologies" (4250.91), "big data" (4022.01), and "qualitative research" (4135.89). These terms emphasize the foundational role of advanced technologies in enabling data-driven decision-making and improving construction efficiency. Their prominence reflects the increasing need to integrate smart technologies to enhance operational performance and safety in construction practices.

Regarding specific technological applications, "digital twin" (1748.36), the "IoT" (716.80), and "blockchain" (869.23) emerge as critical technologies. Both "digital twin" and the "IoT" are crucial for enabling real-time monitoring and predictive analytics, which are pivotal in enhancing safety and optimizing operational efficiency in construction projects. Conversely, "blockchain" technology is fundamental in securing project data and ensuring transparency, safeguarding data integrity, and facilitating informed decision-making.

Furthermore, terms such as "management" (1677.79), "processes" (1482.34), and "applications" (1314.94) highlight the increasing emphasis on efficient project management and streamlined operational workflows. These terms suggest that integrating smart technologies is critical for enhancing safety and optimizing project execution and resource management. The prominence of these terms further underscores the growing recognition that integrated management systems are crucial for refining construction processes and effectively leveraging technological innovations.

Topic 2 (Sustainable Smart Environments Through Technological Integration) explores how the integration of smart technologies can foster sustainable smart environments. Insights from the TF-IDF analysis shed light on the influence of smart technologies in promoting sustainability within the construction industry. The analysis highlights terms such as "smart" (TF-IDF: 445.14), "energy" (889.43), and "carbon" (400.98), underscoring the vital role smart technologies play in boosting energy efficiency and reducing carbon emissions. The emphasis on these terms underlines the mounting importance of sustainability in the construction sector, especially concerning energy management and minimizing environmental impact. This indicates that smart technologies are crucial for promoting sustainable construction practices and developing environmentally friendly building designs.

In the context of urban and environmental impacts, the terms "cities" (709.95) and "environment" (310.65) emerge as significant focal points. These terms denote a growing trend in utilizing smart technologies in urban settings to enhance building performance, improve resource management, and design resilient, sustainable smart cities. The presence of these terms reflects the increasing link between urban development and sustainability objectives, further promoting the application of smart technologies that support green building initiatives and contribute to more sustainable urban planning.

The concepts of "systems" (836.65), "framework" (478.26), and "performance" (835.53) underscore the critical need for comprehensive and effective frameworks to foster sustainability in construction. These concepts advocate for a system-wide, integrated method to the adoption of smart technologies, which is vital for their successful deployment. It is crucial to ensure that technological systems are harmonized with organizational strategies to achieve sustainability goals and yield enduring environmental advantages.

In Topic 3 (Worker-Centered Safety Management Systems and Technological Support), the analysis underscores the increasing emphasis on "worker safety" as a cornerstone of smart construction practices. The TF-IDF analysis identifies several key terms, including

"safety" (TF-IDF: 939.96), "workers" (450.33), and "environment" (310.65). These terms highlight the pivotal role of ensuring "worker protection" and improving onsite conditions, reflecting a paradigm shift in construction safety research. The frequent appearance of these terms indicates a growing recognition of the need to integrate "worker-centric safety measures" into modern construction methodologies, where "worker well-being" and protection are no longer secondary considerations but central to advanced safety management systems.

Further, the analysis identifies critical terms related to technological support, such as "monitoring" (429.21), "support" (507.27), and "collaboration" (223.19). These terms emphasize the importance of leveraging advanced technologies to enhance safety outcomes. For example, "monitoring" technologies, such as wearable sensors and IoT systems, play a crucial role in the early detection of potential hazards, thereby mitigating risks before they escalate into accidents. Similarly, "collaborative tools" facilitate efficient communication and coordination among workers, safety managers, and stakeholders, contributing to organizational efficiency and aligning safety management processes with operational workflows.

Additionally, the analysis highlights the terms "education" (283.44) and "training" (161.88), which underscore the necessity of continuous learning to address dynamic construction risks. As the integration of technologies such as "digital twins" and the IoT continues to evolve, comprehensive "training programs" are essential. These programs equip workers with the knowledge and skills needed to adapt to new tools and safety protocols effectively. For instance, initiatives focused on "digital competency" and "hazard response" empower workers to navigate complex and technologically advanced construction environments, bridging the gap between theoretical frameworks and practical execution.

The results demonstrate the multifaceted challenges and opportunities in developing "worker-centered safety management systems" supported by advanced technologies. The prominence of terms related to "worker safety", "monitoring", and "training" reflects not only the current state of smart construction practices but also the directions for future improvements. The analysis reveals a growing awareness of how real-time "monitoring" and technological "collaboration" can mitigate risks while improving operational efficiency. Simultaneously, the emphasis on "education" and "training" points to the importance of preparing workers for the demands of a rapidly evolving industry.

By analyzing these findings, this study highlights the importance of integrating technological and educational initiatives into comprehensive safety management frameworks. Such integration enables the construction industry to move toward safer and more efficient environments, where "worker safety" and technological innovation work hand in hand. This alignment between research findings and actionable strategies ensures that the frameworks proposed for smart construction safety management remain adaptable and impactful in addressing the challenges of "worker safety" and technological integration.

In conclusion, the TF-IDF analysis across these three research topics reveals a distinct shift toward the integration of smart technologies with traditional construction practices. Specifically, it underscores the heightened focus on worker safety, technology-based support, and continuous training, underscoring the increasing importance of safeguarding workers, optimizing safety management, and cultivating a safety culture in the construction industry. These trends indicate a need for targeted technological solutions and management strategies to meet the distinct challenges of contemporary construction environments.

5. Discussion

In this study, we investigate the application of smart technologies to enhance safety and efficiency in construction and examine the research trends in smart construction safety management, deriving theoretical and practical implications for advancing safety systems

in this domain. To achieve this, we examine the impact of these technologies on safety management at construction sites from multiple dimensions, which allows us to identify three principal research themes.

Firstly, we examine the integration of advanced technologies, such as the Internet of Things (IoT), digital twins, and big data, and demonstrate how their application can significantly enhance the efficiency and safety of construction processes. Secondly, we explore the role of technological integration in fostering a sustainable smart construction environment, focusing on reducing environmental impact and promoting resource efficiency. Thirdly, we emphasize the importance of worker-centered safety management systems, supported by real-time monitoring and technological tools, in developing proactive and adaptive safety frameworks.

To substantiate these research themes, the results of the text frequency analysis offer critical insights into how frequently occurring keywords align with the core objectives of this study. The prominence of terms such as "safety", "risk", "technology", and "sustainability" highlight their central role in the academic discourse on smart construction safety management. For example, the frequent occurrence of the terms "safety" and "risk" underscores the foundational importance of risk management theory, particularly the ISO 31000 framework, in shaping the theoretical basis of the study. Likewise, the recurring appearance of terms like "technology" and "sustainability" reinforces the practical implications of applying advanced technologies to enhance safety and environmental responsibility on construction sites.

These findings reinforce the interrelationship between the text frequency analysis and the identified research themes, demonstrating how the results support the theoretical and practical contributions of this study. By bridging the gap between data-driven insights and conceptual frameworks, this study provides a robust foundation for future research and practical applications in smart construction safety management.

5.1. Theoretical Implications

In this study, we explore how advanced digital technologies can be integrated into existing construction management practices to improve safety in smart construction, highlighting distinct theoretical and practical implications. Specifically, we conduct an in-depth analysis of the roles played by advanced technologies, such as digital twins, the Internet of Things (IoT), and big data, in enhancing the effectiveness of construction safety management systems. Our text frequency analysis reveals the recurring prominence of these terms, underscoring their critical role in shaping modern construction safety practices. By synthesizing insights from academic literature, we identify key applications and trends that provide a solid basis for actionable methodologies.

The emphasis on digital twins highlights their transformative potential in construction safety management. By creating virtual replicas of construction sites, digital twins facilitate real-time monitoring, predictive modeling, and hazard simulation. These capabilities enable testing safety protocols in a controlled virtual environment. For example, Fang et al. [1] developed a digital twin model to manage unsafe construction worker behaviors, utilizing simulated scenarios to proactively identify hazards and modify behavior. Similarly, Azhar [3] demonstrated the utility of digital twins in crane operations, where simulations of load movements helped detect potential collisions and mitigate risks, significantly enhancing operational safety.

IoT technology complements these capabilities by enabling real-time monitoring through wearable sensors. These devices track critical parameters such as worker health, temperature, and air quality, providing immediate alerts to prevent accidents. Bose et al. [2] proposed an IoT-based inventory management system that integrates safety monitoring

into interconnected networks, optimizing resource allocation while mitigating risks. This approach aligns with Salzano et al. [22], who emphasized incorporating IoT and Building Information Modeling (BIM) into comprehensive risk management frameworks to enhance safety and operational efficiency. By combining the predictive capabilities of digital twins with IoT's real-time data acquisition, construction projects can achieve a synergistic integration of theoretical models and practical applications, as noted by Sousa et al. [16].

Our findings also align with ISO 31000-based risk management frameworks, as discussed by Lalonde and Boiral [23]. Extending these frameworks to incorporate digital technologies introduces a new dimension to safety management, significantly enhancing operational efficiency and risk mitigation. For instance, integrating IoT devices with risk management models facilitates proactive hazard identification and response, bridging theoretical principles with practical outcomes.

Additionally, this study emphasizes the necessity of worker-centric safety management systems. Moving from reactive to proactive strategies, these systems prioritize worker engagement through real-time monitoring and continuous education supported by smart technologies. Such an approach builds upon Sousa et al. [16], who advocated risk-based occupational safety management. Incorporating advanced technological capabilities enhances the adaptability and responsiveness of safety systems, making them more effective and inclusive.

In line with these advancements, our analysis also underscores the role of digital technologies in promoting environmental sustainability. By leveraging big data analytics to optimize material usage and minimize waste, we address critical aspects of sustainable construction practices. For instance, Xu et al. [19] highlighted the utility of text-mining techniques to identify safety risks and optimize preventive measures, reinforcing the relevance of predictive analytics in sustainability strategies. These findings resonate with Salzano et al. [22], who discussed integrating BIM into safety and project execution frameworks to simultaneously achieve sustainability and operational excellence.

Lastly, our study identifies emerging trends in smart construction safety management through a text-mining analysis of the existing literature. By employing methodologies such as topic modeling, as outlined by DiMaggio et al. [20], we synthesize complex datasets to uncover actionable insights, including the growing significance of worker-centered systems and sustainability-driven technologies. This analysis provides a nuanced understanding of the evolving landscape of construction safety management.

In conclusion, this study contributes a comprehensive framework for integrating advanced digital technologies into construction safety management. While our findings offer theoretical and practical advancements, future research should validate these insights through empirical studies in diverse construction settings. Such investigations should address challenges related to cost, technical feasibility, and workforce adaptation, as emphasized by Seo and Hong [18]. By resolving these challenges, the construction industry can further optimize the integration of smart technologies, ensuring continued evolution in both theoretical and practical dimensions.

5.2. Practical Implications

This research provides essential foundational insights that can substantially enhance the deployment and effectiveness of smart construction safety systems. It outlines viable strategies for integrating advanced technologies into the construction sector. Notably, the TF-IDF analysis identifies key terms such as "safety", "technology", and "monitoring", which reflect the central role of these concepts in the academic discourse surrounding construction safety. The prominence of terms like "digital twin" and "IoT" in the analysis underscores the critical importance of these technologies in reducing accidents by collecting

and analyzing real-time data directly from construction sites. For instance, Xu et al. [32] developed a method to improve site safety through continuous monitoring facilitated by these smart technologies.

The high frequency of related terms such as "monitoring" (TF-IDF: 429.21) and "support" (507.27) highlights the growing academic focus on technologies that enable real-time hazard detection and risk mitigation. TF-IDF, as a statistical measure of word relevance, indicates that these terms are highly significant within the context of smart construction safety. The increased frequency of "monitoring" emphasizes the increasing recognition of real-time monitoring systems as a key component of modern safety management, supporting proactive risk management approaches. Similarly, the term "support" refers to the technological tools and systems that are critical for enabling workers and safety managers to proactively identify and mitigate potential hazards before they escalate into accidents. Such innovations, as evidenced by the TF-IDF analysis, hold great promise for improving safety outcomes at construction sites, offering a robust framework for businesses to upgrade their safety protocols. By integrating these technologies, as indicated by the dominant terms in the analysis, this research aligns with the growing trend of incorporating data-driven approaches to safety management in the construction industry.

Second, this study emphasizes the effectiveness of smart construction safety technologies in facilitating a more proactive approach to managing worker safety. Unlike traditional safety management systems, which predominantly rely on reactive measures, the real-time monitoring system proposed in this study is designed to proactively identify and mitigate potential risks. This shift is vital for enhancing on-site safety for workers and provides construction companies with more efficient and effective means to address safety challenges. Furthermore, integrating advanced technologies into educational and training systems can significantly improve safety awareness among workers, potentially leading to a further reduction in workplace accidents. Li et al. [14] investigated methods to enhance safety education effectiveness through the application of these smart technologies. By incorporating these technologies into safety management frameworks, this study aligns with risk management theories, such as those based on the ISO 31000 framework [15,25], which underscore the importance of proactive hazard identification and risk mitigation.

Third, this study underscores the vital role of smart technology in promoting a sustainable construction environment. The concepts of smart buildings and smart cities not only improve the efficiency of construction sites but also present opportunities to contribute to sustainable societal development through integration with extensive urban and environmental management systems. A significant practical challenge for the construction industry lies in advancing environmental sustainability via improved energy efficiency and reduced carbon emissions. Righi et al. [31] examined how smart construction can aid in the development of sustainable cities and environmental management, a perspective in alignment with this study's findings [31].

In alignment with these practical applications, the theoretical implications of this study further expand on the evolving role of technology within safety and sustainability frameworks. By illustrating how smart technologies like the IoT and digital twins can integrate into and enhance traditional risk management models, this research bridges the gap between conceptual theories and real-world implementation. Notably, the findings reaffirm the role of technology-driven systems in advancing environmental and worker safety goals, as advocated by studies like that of Salzano et al. [22], which emphasize integrating Building Information Modeling (BIM) into sustainable and safety-conscious construction practices [5,36,37]. This alignment not only validates the theoretical models explored but also paves the way for their adaptation and scalability across diverse construction contexts.

6. Conclusions

This study presents an in-depth examination of the integration of advanced smart construction technologies—specifically digital twins, the Internet of Things (IoT), and big data—within safety management systems in construction environments. Unlike prior research that has predominantly focused on conceptual frameworks or theoretical models, this study provides a pragmatic, actionable methodology for the real-world deployment of these technologies, effectively bridging the gap between theoretical constructs and practical implementation in smart construction safety management.

The TF-IDF analysis conducted herein reveals the prominence of key terms such as "safety", "technology", and "risk management", which appear with significant frequency, underscoring their central role in the discourse surrounding the impact of technological advancements on safety management systems. The frequent appearance of terms like "digital twin", "IoT", and "big data" further highlights the critical function these technologies serve in enhancing safety management. These technologies facilitate real-time monitoring, predictive risk management, and resource optimization, enabling a proactive approach to identifying and mitigating risks, thus improving operational performance and safety outcomes. The prominence of these terms within the text underscores the growing recognition of the value of integrating smart technologies as pivotal mechanisms in reducing safety hazards and advancing safety management theories.

Moreover, this study underscores the dual benefits of smart technologies in advancing environmental sustainability within construction processes. By contributing to enhanced energy efficiency, reduced carbon emissions, and the promotion of sustainable practices, these technologies play a significant role in both improving safety outcomes and fostering environmental stewardship. The consistent emergence of terms such as "environment" and "sustainability" in the TF-IDF analysis further reflects the increasing integration of environmental considerations into smart construction practices. These findings are consistent with current research that advocates for the widespread adoption of smart technologies, emphasizing not only their potential to enhance safety but also their ability to promote resource efficiency and environmental responsibility in construction practices.

Another key aspect of our research is the shift toward worker-centered safety management systems, moving away from traditional, reactive approaches. We illustrate how real-time monitoring, combined with continuous education and training, creates a dynamic, proactive safety management system. Technologies such as the IoT and wearable devices enable responsive safety frameworks that not only address immediate hazards but also promote a culture of safety awareness within the workforce.

Despite the value of this research, it is essential to acknowledge its limitations, which also point to areas for future exploration. Although we developed a theoretical model for incorporating smart technologies into construction safety management, the lack of empirical validation across various construction environments remains a significant limitation. Differences in project types, site conditions, and organizational structures may influence the applicability and impact of specific smart technologies. Future research should involve broader case studies, particularly across different types of construction projects, to refine the proposed methodologies and develop more universally applicable guidelines for implementation.

Additionally, further investigations are needed to address the practical barriers hindering the widespread adoption of smart technologies, particularly in terms of financial constraints, technical feasibility, and workforce acceptance. While the benefits of these technologies are clear, obstacles such as high implementation costs, technical complexities, and resistance from workers unfamiliar with these innovations must be examined in greater

detail. Understanding these factors is critical to overcoming the challenges to adopt and facilitate the smooth integration of smart technologies into the construction industry.

Future studies should also focus on developing comprehensive models that integrate smart technologies into core construction processes, such as safety management, project scheduling, and resource allocation. Methodologies that quantitatively assess the impact of these technologies on safety and project performance are needed, as well as the development of standardized guidelines for their scalability, cost-effectiveness, and ease of deployment across diverse construction settings. These efforts are vital for creating a clear roadmap for the successful integration of smart technologies, providing valuable insights for industry practitioners and policymakers working to improve safety and sustainability in the construction sector.

In conclusion, this study offers critical insights into the integration of smart technologies into construction safety management systems. By providing a practical approach for deploying advanced technologies like digital twins, the IoT, and big data, our research highlights their potential to enhance safety, improve operational efficiency, and promote sustainability. While this study advances theoretical understanding, it also suggests actionable strategies for real-world implementation. However, due to the lack of empirical validation, future research should aim to incorporate case studies or field investigations to substantiate the effectiveness and broader applicability of these technologies across diverse construction environments. This would ensure the robustness of the findings and further clarify how smart technologies can be effectively integrated into safety management systems within the industry.

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